

Notes: Subrahmanyam and Titman (JF, 1999)

The Going-Public Decision and the Development of Financial Markets

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Model and primitives	4
3.1	Entrepreneur and production technology	4
3.2	Two financing regimes	4
3.3	Information structure	5
3.4	Security market and liquidity	5
4	Public financing	5
4.1	Equilibrium information aggregation	5
4.2	Serendipitous information and costly information acquisition	6
4.3	Resource allocation under public financing	6
5	Private financing	6
6	Public versus private financing	7
6.1	Main financing trade-off	7
6.2	Crowding out	7
6.3	Comparative statics	8
6.4	Limiting results	8
7	Financial-market development and externalities	8
7.1	Positive externalities from going public	8
7.2	Multiple equilibria	8
7.3	Market size and snowballing	9
7.4	Negative externalities	9
7.5	Global markets and home bias	9
8	Interpretation and identification	9
8.1	What the model can explain	9
8.2	Key assumptions	10

8.3	Why public prices matter	10
8.4	Relation to bank- versus market-based systems	10
9	Tables and figures	11
9.1	Table I: An example of crowding out	11
9.2	Figure 1: Trade-offs in public versus private financing	11
9.3	Figure 2: The benefit to public financing as a function of market size	12
10	Empirical implications	13
11	Short summary	13
12	Conclusion	14
12.1	Core conclusion	14
12.2	Economic intuition	14
12.3	Financial-market development	14
12.4	Implications	15
12.5	Limitations	15
12.6	Takeaway	15

1 Big picture

- **Research question.** Why do some firms finance projects through public equity markets while others remain privately financed, and how does this choice relate to the development of financial markets?
- **Main idea.** Public markets can aggregate information from many investors. This information can improve real investment decisions, but public trading also creates liquidity costs because informed traders trade against liquidity traders.
- **Core trade-off.** Private financing avoids the adverse-selection and duplication costs of public trading, but it relies on a private financier's more limited information set. Public financing is attractive when the stock market can aggregate diverse or serendipitously obtained information that a private financier would not have.
- **Information types.** The paper distinguishes information acquired through costly analysis from information obtained serendipitously in investors' ordinary activities. Costly information may be duplicated by analysts, while serendipitous information is naturally dispersed across many investors.
- **Going-public decision.** An entrepreneur chooses public financing when the value created by better stock-price information exceeds the liquidity cost and any duplication of information acquisition.
- **Market-development feedback.** The value of being public depends on the size and liquidity of the public market. A larger public market attracts more active investors, improves price informativeness, and can make it more attractive for more firms to go public.
- **Multiple equilibria.** Because a firm's decision to go public improves the market environment for other firms, firms may ignore a positive externality. Economies can be stuck in a bad equilibrium with too few public firms or move to a good equilibrium with many public firms.
- **Economic takeaway.** Financial-market development is partly a coordination problem. Public equity markets are valuable not only because they supply capital, but because they aggregate information that guides real resource allocation.

2 Incremental contribution of the paper

- **From financing cost to information production.** Standard explanations for going public often emphasize diversification, liquidity, or risk sharing. This paper emphasizes how public markets generate and aggregate information used in real decisions.
- **A theory of stock prices guiding real investment.** The paper links market price informativeness to the entrepreneur's choice of capital allocation in a growth opportunity.
- **Public versus private financing as information regimes.** Public financing is modeled as a decentralized information market, while private financing is modeled as a delegated information-collection regime.

- **Serendipitous information.** A distinctive contribution is to formalize the idea that investors may obtain useful information without intentionally researching the firm. This information is hard for a private financier to reproduce.
- **Crowding-out channel.** The paper shows that more serendipitous information does not always monotonically improve public markets. It can crowd out costly information acquisition and, in some parameter regions, reduce the attractiveness of going public.
- **Market-size externality.** The model explains why the value of public financing depends on the number of firms already traded publicly. This creates path dependence in financial market development.
- **Institutional and policy implications.** The theory gives a mechanism through which lower trading costs, lower information costs, better disclosure, and privatization programs can jump-start public equity markets.
- **Empirical implications.** The model rationalizes IPO clustering, spinoffs, LBOs followed by re-IPOs, the rise of public markets after trading-cost reductions, and differences across financial systems such as the United States versus Germany.

3 Model and primitives

3.1 Entrepreneur and production technology

The entrepreneur has assets in place and a growth opportunity. The payoff on assets in place is

$$F + \delta + \theta,$$

where F is known ex ante, and δ and θ are independent zero-mean shocks.

The growth opportunity requires a capital choice K and pays

$$G + K(\delta + \theta) - \frac{1}{2}K^2.$$

The entrepreneur chooses K after receiving information from either a private financier or the public stock price. The optimal investment rule is therefore

$$K = E(\delta + \theta \mid \text{available information}).$$

Interpretation. Better information about $\delta + \theta$ raises firm value because the entrepreneur can choose K closer to the true investment opportunity.

3.2 Two financing regimes

The entrepreneur can obtain initial financing in one of two ways.

- **Public financing.** The entrepreneur issues claims that trade in a public security market. The stock price aggregates information from informed traders and is observed by the entrepreneur before choosing investment.

- **Private financing.** The entrepreneur sells the firm to a private financier, such as a conglomerate, venture capitalist, or financing company. The private financier collects information directly and guides investment.

The private financier is assumed to be well diversified, so its own shares are more liquid than the entrepreneur’s stand-alone public stock would be. This gives private financing a liquidity advantage.

3.3 Information structure

The paper separates two information sources.

- **Costly information.** Agents can pay cost c to obtain a signal about δ . In the public market, M traders choose to collect this costly information. A signal has a common noise component and an idiosyncratic noise component, so costly information can be either highly correlated or diverse.
- **Serendipitous information.** Some investors receive information about θ by chance, without paying a research cost. In the public market, N traders receive such signals. The private financier is assumed not to observe this information.

Economic interpretation. Costly information is like analyst research about the firm. Serendipitous information is like a retailer, customer, supplier, or analyst of another company accidentally learning something about demand for the firm’s product.

3.4 Security market and liquidity

In the public market, informed traders submit orders based on their signals, liquidity traders trade for noninformational reasons, and competitive market makers set price from total order flow:

$$P = F + \lambda Q,$$

where Q is total order flow and λ is the price-impact or illiquidity parameter.

The stock price has two effects.

- It conveys information to the entrepreneur and improves the investment decision.
- It creates a liquidity cost because liquidity traders lose to informed traders, and investors require compensation for bearing this adverse-selection cost.

Interpretation. Public markets are useful because price is informative, but costly because the same information makes secondary-market trading less liquid.

4 Public financing

4.1 Equilibrium information aggregation

In public financing, the stock price combines information from two groups: costly informed traders and serendipitously informed traders. The paper summarizes the contribution of each group with terms analogous to T_M and T_N .

- T_M captures the effect of costly information traders on price informativeness and liquidity.
- T_N captures the effect of serendipitously informed traders.
- The number of costly informed traders M is endogenous because traders enter until expected trading profits equal the information cost c .

A key result is that adding more informed traders has two opposing effects.

- **Competition effect.** More informed traders compete away informational rents and make the market more liquid.
- **Information effect.** If different informed traders have genuinely different signals, more traders bring more information into the price.

Economic message. The informational benefit of a public market depends not simply on the number of analysts, but on whether their information is diverse or duplicative.

4.2 Serendipitous information and costly information acquisition

Increasing the number N of serendipitously informed traders can either encourage or discourage costly information acquisition.

- If N is sufficiently large, additional serendipitous traders increase liquidity. This raises the profitability of costly information acquisition and increases M .
- If N is small and serendipitous signals are noisy, additional serendipitous trading can reduce liquidity and lower the incentive to collect costly information.

This is the paper's crowding-out mechanism: serendipitous information can crowd out costly information acquisition when the market is not yet deep enough.

4.3 Resource allocation under public financing

The entrepreneur uses the price to infer $\delta + \theta$ and choose K . Substituting the optimal K into the payoff of the growth opportunity shows that expected firm value rises with the precision of price information.

The value of public financing is therefore the value of assets in place plus the expected value of the growth opportunity, minus the liquidity cost from informed trading.

Interpretation. Public financing is not just a way to raise capital. It is a real resource-allocation mechanism because price information affects investment.

5 Private financing

Under private financing, the private financier buys the firm and collects a costly signal. The private financier does not generate the public-market liquidity cost and avoids duplicative analyst effort, but it observes a smaller and less diverse information set.

- Private financing is especially attractive when costly information is highly correlated across agents. In that case, public analysts duplicate one another's effort.
- Private financing is also attractive when information relevant to investment requires focused due diligence rather than dispersed investor knowledge.
- Private financing is less attractive when many dispersed investors have different pieces of valuable information.

Economic message. A private financier is an efficient centralized information producer, but it cannot aggregate broad, dispersed, serendipitous information as well as a large public market.

6 Public versus private financing

6.1 Main financing trade-off

The entrepreneur chooses public financing if

$$\begin{aligned} &\text{information benefit of the public price} - \text{public-market liquidity cost} \\ &> \text{information benefit of private financing} - \text{private information cost.} \end{aligned}$$

The sign of this inequality depends on the information environment.

- If costly information is perfectly correlated and there is no serendipitous information, private financing dominates. Public analysts duplicate one another, and liquidity costs make public financing inferior.
- If costly information is diverse and cheap, public financing can dominate because the market aggregates many different signals.
- If serendipitous information is valuable and sufficiently widespread, public financing becomes more attractive because this information is not easily replicated by a private financier.

6.2 Crowding out

The paper's numerical example shows that the value of public financing need not be monotonic in the amount of serendipitous information.

In Table I, public financing is attractive when N is very small or sufficiently large, but private financing is preferred for intermediate values of N . The mechanism is:

- an increase in N initially lowers the number M of costly informed traders;
- if costly information is diverse and important, this reduction in M worsens price informativeness;
- when N becomes large enough, the direct benefit of serendipitous information dominates and public financing becomes attractive again.

Interpretation. Public markets can suffer from a nonmonotonic information problem. More free information can be helpful, but only if it does not undermine costly information production too much.

6.3 Comparative statics

The model yields several comparative statics.

- Public financing is more attractive when the number of serendipitously informed traders is high.
- Public financing is more attractive when serendipitous information is precise and valuable.
- Public financing is more attractive when costly information is diverse and cheap to acquire.
- Public financing is less attractive when costly information is expensive or when public-market trading is highly illiquid.
- Public financing is less attractive when firm-relevant information is mostly internal, hard to verify, and best obtained through concentrated due diligence.

6.4 Limiting results

The paper emphasizes two limiting cases.

- As the number of serendipitously informed traders grows large, public financing becomes more attractive.
- As the cost of information acquisition goes to zero, public financing can dominate when costly signals are diverse, because the public market can aggregate many signals.

Economic message. Deep, liquid, low-information-cost markets tilt firms toward public financing. Thin or duplicative-information environments tilt firms toward private financing.

7 Financial-market development and externalities

7.1 Positive externalities from going public

When one firm goes public, it can improve the public market for other firms.

- More public firms increase the payoff from becoming an active stock-market investor.
- More active investors increase the amount of serendipitous information in the market.
- A larger investor base can increase liquidity and induce more costly information acquisition.
- Better liquidity and price informativeness make it more attractive for additional firms to go public.

Interpretation. A firm's private going-public decision creates a public good: a more liquid and informative market.

7.2 Multiple equilibria

Because firms ignore their contribution to overall market depth and investor participation, the economy can have multiple equilibria.

- **Bad equilibrium.** Few firms go public, few investors become active, prices are not very informative, and firms prefer private financing.
- **Good equilibrium.** Many firms go public, many investors become active, prices are informative, and firms prefer public financing.

Firm values are higher in the good equilibrium. The problem is that a single firm does not internalize the effect of its listing on the whole market.

7.3 Market size and snowballing

Figure 2 illustrates the market-size effect. As the number of public firms rises, the relative benefit of public financing rises as well. This can create a snowballing process:

more listed firms $\Rightarrow$ more active investors $\Rightarrow$ more informative prices $\Rightarrow$ more firms want to list.

Policy implication. Small reductions in trading costs, information costs, or disclosure frictions can have large effects if they push the economy past a critical mass.

7.4 Negative externalities

The paper also allows for negative externalities. If serendipitous trading crowds out costly information acquisition, an additional public firm can reduce price informativeness for other firms. In that case, too many firms may go public.

Interpretation. The sign of the externality depends on whether additional market participation mainly improves information aggregation or mainly undermines costly research.

7.5 Global markets and home bias

Globalization does not fully eliminate the market-development problem. Investors may be more likely to receive serendipitous information about domestic firms, firms with well-known brands, exporters, or firms with a broad international presence.

Implication. International public markets are especially useful for firms whose products and information flows are global. Firms with localized products may still depend more on domestic markets or private financing.

8 Interpretation and identification

8.1 What the model can explain

The paper is a theoretical model, so it does not estimate causal effects directly. It provides a mechanism that can explain several observed patterns:

- why some firms go public and others remain private;

- why IPOs cluster by industry;
- why firms may go private before restructuring and later return to public markets;
- why spinoffs and public equity issuance may rise when public markets become more liquid;
- why stock market development can be path dependent across countries.

8.2 Key assumptions

The main assumptions are useful but stylized.

- The private financier is assumed to be more liquid than the stand-alone public firm.
- The private financier receives costly information but not serendipitous information.
- Public-market prices affect real investment decisions.
- Managers are assumed to act in investors' interests in the baseline model.
- Investors are risk neutral, so risk-sharing motives are intentionally set aside.

8.3 Why public prices matter

The model only requires a link between stock prices and real decisions. This link can arise because managers learn from prices, because prices affect the cost of raising future capital, or because price-based compensation and monitoring affect managerial incentives.

Interpretation. The model's core logic does not depend on a single microfoundation for why prices matter. It depends on the broader idea that more informative prices improve corporate decisions.

8.4 Relation to bank- versus market-based systems

The model gives a way to compare financial systems.

- A private-financier system may be better at producing costly, internal, due-diligence information.
- A public-market system may be better at aggregating dispersed, external, and product-market information.
- Which system is better depends on what kind of information is most important for real resource allocation.

Economic message. Public markets and private financiers are not simply substitutes for funding. They are different information technologies.

9 Tables and figures

9.1 Table I: An example of crowding out

Read:

- Table I reports a numerical example in which N is the number of serendipitously informed traders, M is the number of costly informed traders, and Δp is the difference between firm value under public and private financing.
- As N rises from 0 to 6, M falls from 19.44 to 15.64, showing that serendipitous information can crowd out costly information acquisition.
- The value difference Δp is positive at $N = 0$ and $N = 1$, negative for $N = 2$ through $N = 6$, and positive again from $N = 7$ onward.
- The example shows that public financing can be unattractive at intermediate levels of serendipitous information but attractive again when the public market becomes sufficiently informative.

Economic message: The benefit of public financing is nonmonotonic because more serendipitous information can reduce costly information acquisition before the direct information benefit becomes large enough.

Table I
An Example of Crowding Out

The numbers of serendipitously informed traders and traders with costly information are respectively denoted by N and M , and Δp denotes the difference between firm values under public and private financing. Parameter values are as follows: the variance of costly information $v_s = 1.5$, the uncorrelated noise variance in costly information $v_e = 3.5$, the common noise variance in costly information $v_\xi = 0.23$, the variance of serendipitous information $v_\theta = 1.2$, the noise variance in serendipitous information $v_\eta = 10$, the variance of liquidity shocks $v_z = 20$, and the cost of information $c = 0.08$.

N	M	Δp
0	19.44	0.019
1	18.24	0.001
2	17.29	-0.011
3	16.66	-0.016
4	16.21	-0.015
5	15.89	-0.011
6	15.64	-0.005
7	15.46	0.003
10	15.11	0.028
20	14.90	0.117
50	15.59	0.284

Table I: An example of crowding out.

9.2 Figure 1: Trade-offs in public versus private financing

Read:

- Figure 1 plots the difference between firm value under public financing and private financing as key parameters change.
- Panel A shows the simple comparative statics: public financing becomes more attractive when there are more serendipitously informed traders, when serendipitous information is more valuable, when costly information is more diverse, and when the cost of information is lower.

- Panel B shows that some relationships can be nonmonotonic, especially when serendipitous information crowds out costly information acquisition.
- The vertical axis is the difference in firm value under public and private financing. Values above zero favor public financing; values below zero favor private financing.

Economic message: The figures summarize the core trade-off: public markets are most valuable when they aggregate many useful, diverse pieces of information at low liquidity and information costs.

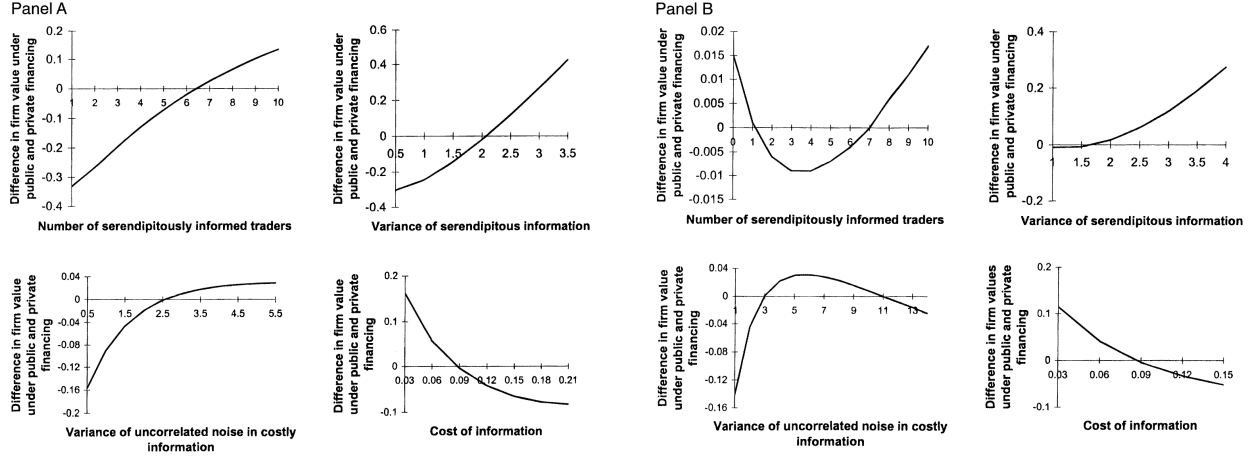


Figure 1. Trade-offs in public versus private financing. The parameter values for Panel A (Panel B) are as follows: the variance of costly information $v_s = 2$ (2), the uncorrelated noise variance in costly information $v_c = 2$ (2.7), the common noise variance in costly information $v_e = 0.1$ (0.08), the variance of serendipitous information $v_\theta = 2$ (1), the noise variance in serendipitous information $v_\eta = 3$ (10), the variance of liquidity shocks $v_z = 1$ (10.7), and the cost of information $c = 0.1$ (0.095).

Figure 1. Continued.

Figure 1, Panel B: Continued comparative statics and nonmonotonic effects.

Figure 1, Panel A: Trade-offs in public versus private financing.

9.3 Figure 2: The benefit to public financing as a function of market size

Read:

- Figure 2 plots the value advantage of public financing as the number of public firms increases.
- In the simulation, the value of public financing rises with the size of the market.
- The benefit is negative when the market is very small, but becomes positive as more firms list.
- This provides the visual foundation for the paper's multiple-equilibria and snowballing arguments.

Economic message: Public markets become more valuable as they grow. This creates path dependence and helps explain why some economies may have underdeveloped stock markets even when a more developed market would be better.

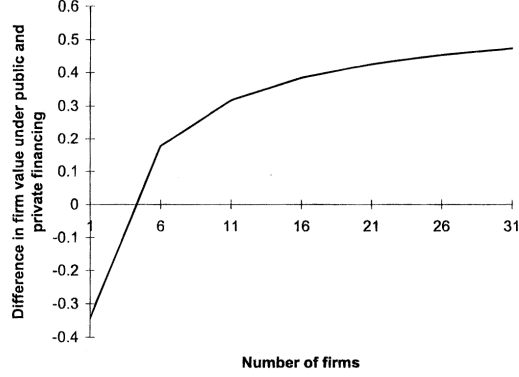


Figure 2. The benefit to public financing as a function of market size. The parameter values are the same as those for Figure 1, Panel A, except that the number of potential serendipitously informed traders is endogenous and the cost of becoming an active investor $C' = 0.3$.

Figure 2: The benefit to public financing as a function of market size.

10 Empirical implications

The paper is theoretical, but it derives several testable implications.

- **Product-market breadth.** Firms with broader product markets and greater demand uncertainty should be more likely to go public because many investors may receive useful demand information.
- **IPO clustering.** IPOs should cluster by industry when investors and analysts following related firms obtain information that is also useful for valuing new firms in the same industry.
- **Public-to-private transactions.** Firms that need internal restructuring may prefer private financing because private financiers are better at concentrated due diligence.
- **Spinoffs.** As public markets become more liquid and informative, conglomerates may spin off divisions because public markets can allocate capital and information better than internal capital markets.
- **Trading-cost reductions.** Lower brokerage commissions and lower information costs should increase IPO activity and make public markets more attractive.
- **Emerging markets.** Privatizations, disclosure improvements, and lower investor participation costs can push markets toward a higher-listing equilibrium.
- **Home bias.** Investors should be more willing to hold and trade firms about which they receive local or serendipitous information, especially firms with domestic or familiar product-market exposure.

11 Short summary

- The paper studies the choice between public and private financing.
- The central mechanism is information aggregation: public markets aggregate dispersed investor information and can guide real investment.

- Public financing has a cost because informed trading reduces liquidity and raises the cost of public capital.
- Private financing avoids public-market liquidity costs and duplicated research, but it uses a narrower information set.
- Public financing is attractive when costly information is diverse and cheap, or when serendipitous information is valuable and widespread.
- Serendipitous information can crowd out costly information, so more free information does not always monotonically increase the value of public financing.
- A firm's decision to go public creates externalities by increasing market size, investor participation, liquidity, and price informativeness.
- These externalities can generate multiple equilibria: a small, illiquid public market equilibrium and a large, liquid public market equilibrium.
- The model helps explain IPO clustering, spinoffs, LBOs, emerging-market development, and the role of lower trading and information costs.
- The main takeaway is that public markets are valuable as information systems, not merely as pools of capital.

12 Conclusion

12.1 Core conclusion

Subrahmanyam and Titman show that the going-public decision depends on how different financing regimes produce and aggregate information. Public markets are valuable when stock prices aggregate dispersed information that improves real investment decisions. Private financing is valuable when information is best collected through focused, costly due diligence and when public-market liquidity costs are large.

12.2 Economic intuition

A public firm benefits from the eyes and ears of many investors. Some investors actively research the firm; others accidentally learn useful information through their normal activities. The stock price aggregates these signals and can help the firm choose better investments. But informed trading makes the market less liquid, and analysts may duplicate one another's effort. The going-public decision balances these forces.

12.3 Financial-market development

The model's most important macro implication is that financial markets can be path dependent. When few firms are public, few investors participate and prices are not very informative. When many firms are public, investors participate, prices become informative, and listing becomes more attractive. Because firms do not internalize this externality, economies may remain stuck with too few public firms.

12.4 Implications

The findings imply that policies reducing investor participation costs, trading costs, and information acquisition costs can have large effects. Disclosure requirements, brokerage-cost reductions, and privatizations can make public markets more attractive and potentially move an economy from a low-market-development equilibrium to a high-market-development equilibrium.

12.5 Limitations

The model is stylized. It abstracts from many incentive problems, assumes risk neutrality, gives the private financier no serendipitous information, and does not estimate the size of the effects empirically. These simplifications make the mechanism transparent but mean that the model should be read as a theory of information externalities rather than a complete account of all going-public decisions.

12.6 Takeaway

The enduring takeaway is that public markets are information platforms. Firms go public not only to raise capital, but to use market prices as signals for real decisions. Because each listing can improve the information environment for other firms, financial market development can display coordination failures, multiple equilibria, and snowballing growth.